Foundation models

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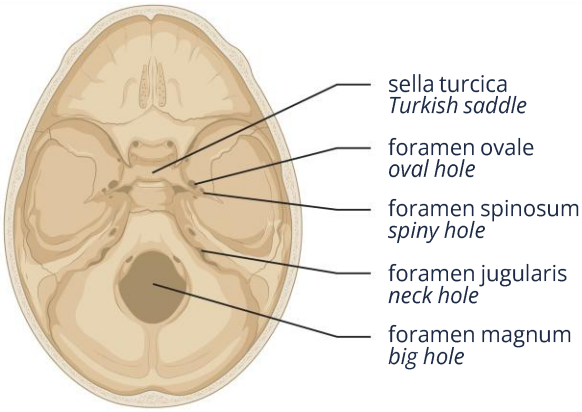
Machine learning methods in biomedical data
Academic year 2025-2026

It is easier to learn patterns if you know the language



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Recognizing patterns is much easier when one already understands the underlying “language” of the data. In anatomy, knowing the Latin language makes it easier to memorize the structure terminology, as the terms are now interpretable. The same general principle applies in machine learning: models perform better when they first learn the structure and conventions of the data domain before tackling a specific downstream prediction task.

Foundation models can learn general sequence 'language' from a large corpus of text

Learn the language from a general task
Predict the missing word

A sentence _____ words.

↓

A sentence eats words. ✗

A sentence contains words. ✓

Predict the missing nucleotides

ATACG___GCTAGTCC

↓

ATACGCCGGCTAGTCC ✗

ATACGTTAGCTAGTCC ✓

No exact labels required
Usually, a lot of data available

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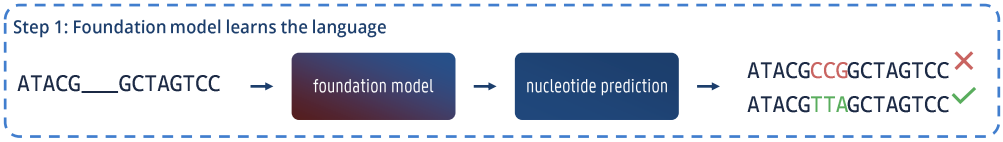
Foundation models are typically trained on very large amounts of data using a general learning objective rather than a narrowly defined task. One common strategy is to hide part of the input and ask the model to predict the missing token, word, or nucleotide. Because the supervision comes from the data itself (i.e., it is **self-supervised**), this approach usually does not require exact manual labels and can exploit the large unlabeled datasets that are common in biology and medicine.

Pretraining & fine-tuning turns one model into many predictors



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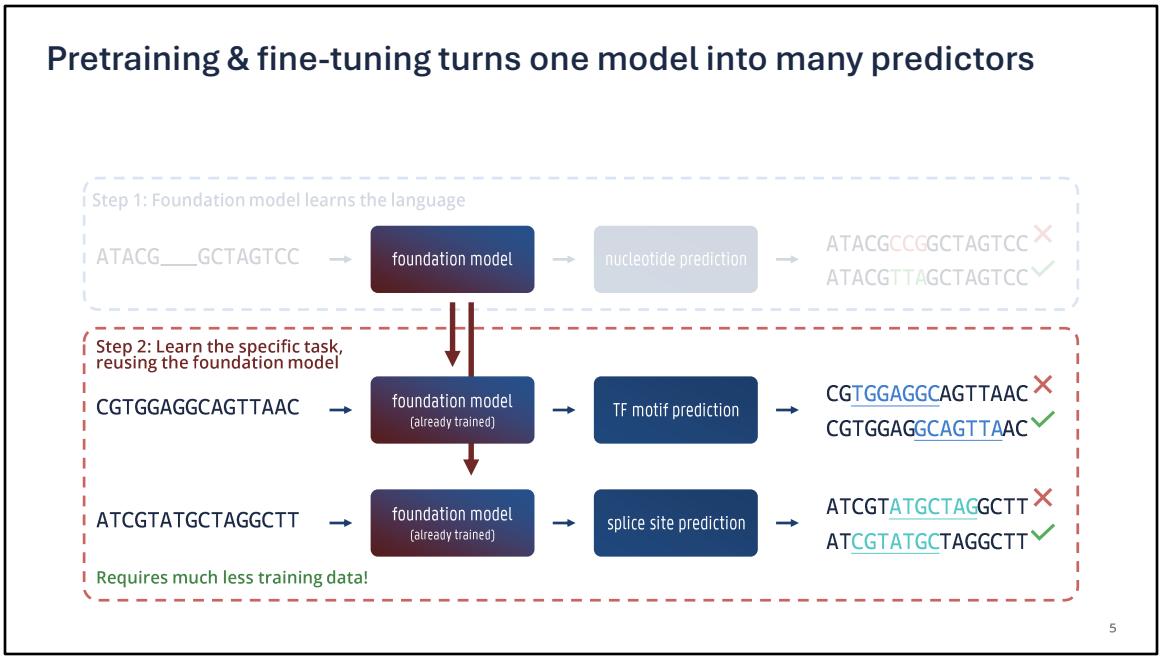
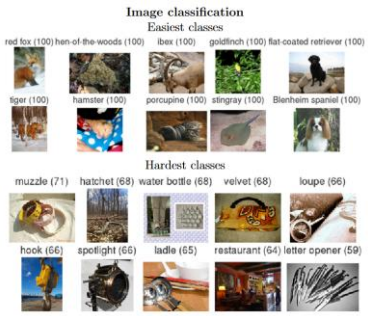
The key idea behind foundation models is that one broadly trained model can later be adapted to many different downstream tasks. In the **pretraining** phase, the model learns general patterns and structure from a large corpus. In the **fine-tuning** phase, that pretrained knowledge is reused and specialized for a particular prediction problem.



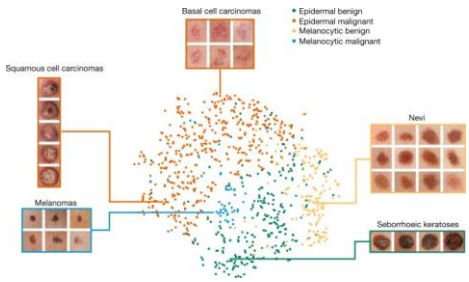
Once a foundation model has learned the general “language” of a data type, it can be reused for several more specific tasks with much less task-specific training data. Instead of training a separate model from scratch for each application, the pretrained model acts as a strong starting point that already captures useful domain structure. This is especially attractive in biomedicine, where labeled datasets are often limited or expensive to obtain.

Foundation models can be highly generic, while the subsequent finetuning task can be very domain-specific

Inception v3, as foundation model
trained on >1m images of 1k object categories



Finetuned for skin lesion classification
finetuned with 127,463 images of skin lesions



Esteva, A. et al. Dermatologist-level classification of skin cancer with deep neural networks. Nature 542, 115–118 (2017). doi:10.1038/nature21056

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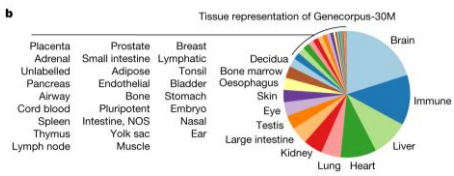
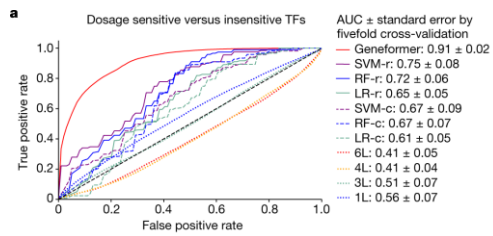
A foundation model does not always need to be trained on domain-specific data from the start. In some cases, a very general model trained on a broad and diverse dataset can still provide useful representations that transfer well to a much more specialized task. The skin-lesion example illustrates this idea: a model pretrained on generic natural images can still serve as a useful starting point for medical image classification after task-specific fine-tuning.

Foundation models can also be more domain-specific, but exploit large corpuses of unlabeled data

Pretraining on transcriptomics data
~30 million single cell transcriptome samples



Finetuned for copy number variation gene dosage sensitivity
10.000 transcription factors with known sensitivity



Theodoris, C.V. et al. Transfer learning enables predictions in network biology. Nature 618, 616–624 (2023). doi:10.1038/s41586-023-06139-9

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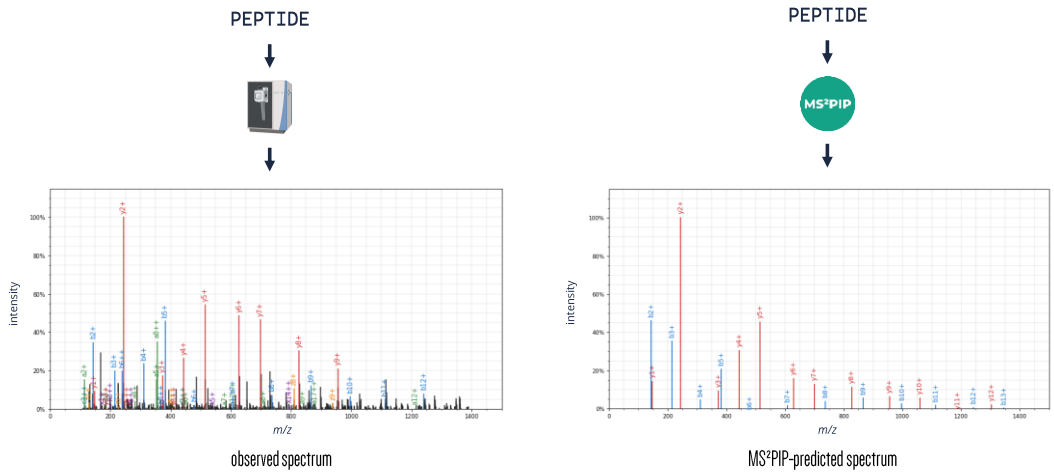
Foundation models can also be trained directly within a specific domain when sufficiently large unlabeled datasets are available. In that case, pretraining can capture more domain-relevant structure than a generic model would. The transcriptomics example illustrates how large-scale pretraining on single-cell data can later support a much more specific biological prediction task with comparatively limited labeled examples.

An example from CompOmics:
Machine learning can predict peptide fragmentation spectra



Ralf Gabriels et al. Nucleic Acids Res. (2019) doi:10.1093/nar/gkz299

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In the next slides follows an extended foundation model example from within the CompOmics group. In the context of mass spectrometry-based proteomics, machine learning models can be trained to predict the fragmentation spectra for any given peptide. This is highly useful in peptide identification workflows.

Amino acids can be directly encoded in various ways

Manual feature engineering

ACDTK

	A	C	D	T	K
Basi	37	35	59	129	191
Heli	68	23	33	29	32
Hydr	51	75	25	35	0
pI	32	23	0	27	69

one-hot encoding

ACDTK

	A	C	D	T	K
A?	1	0	0	0	0
C?	0	1	0	0	0
D?	0	0	1	0	0
…	…	…	…	…	…
T?	0	0	0	1	0
…	…	…	…	…	…
K?	0	0	0	0	1
…	…	…	…	…	…

learned embedding

ACDTK

A	C	D	T	K
0.73	0.14	0.88	0.46	0.21
0.57	0.95	0.32	0.69	0.04
0.81	0.27	0.53	0.10	0.76
0.38	0.62	0.17	0.84	0.49
0.92	0.35	0.58	0.07	0.64
0.11	0.79	0.43	0.96	0.28
0.67	0.24	0.85	0.51	0.13
0.40	0.73	0.19	0.60	0.88

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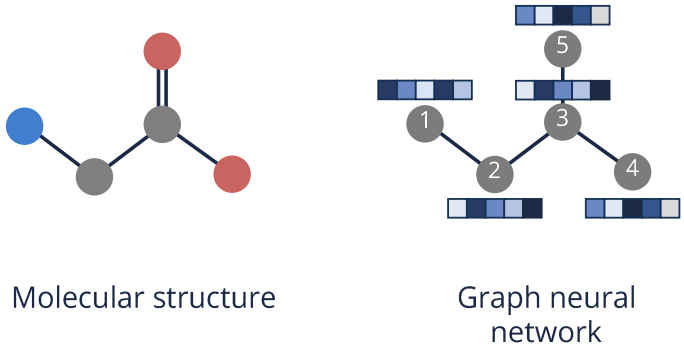
As seen in the lecture on recurrent neural networks, before a model can learn from amino-acid sequences, those symbols must first be converted into numbers. This can be done through **manual feature engineering**, **one-hot encoding**, or **learned embeddings**. These encodings differ in how much prior biological knowledge they inject and in how flexibly they allow the model to learn useful similarities between residues.

A novel encoding system for molecules is a graph neural network, where molecular structure defines network topology



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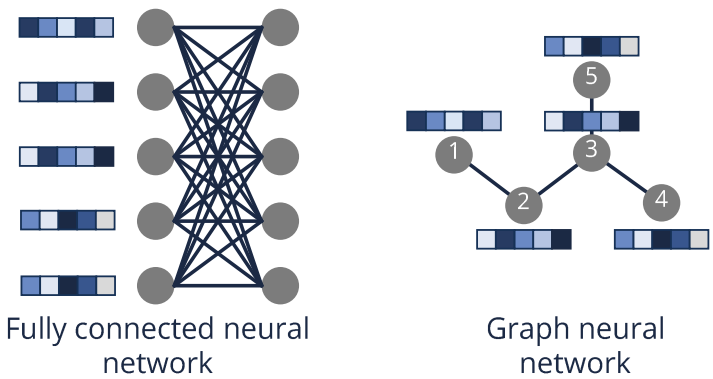
S Kearnes et al. J Comput Aided Mol Des. (2016) doi:10.1007/s10822-016-9938-8; A Kensert et al. *ArXiv* (2023) doi:10.48550/arXiv.2208.09944

Not all inputs are best represented as simple sequences or vectors. For molecules, the connectivity between atoms is itself biologically meaningful, so it can be advantageous to represent the molecule as a graph. A graph neural network (GNN) then processes that structure directly, allowing the model to use relational information that would be difficult to capture with a standard fully connected representation. As amino acids are also molecules, we can extend this GNN-based strategy to peptide sequences.

A novel encoding system for molecules is a graph neural network, where molecular structure defines network topology



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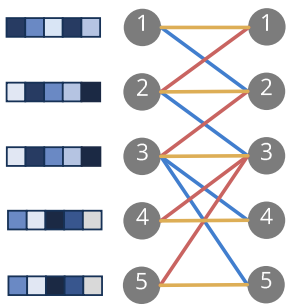
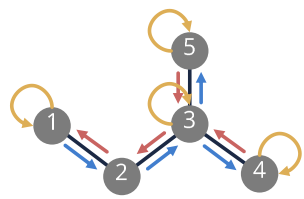
In a standard fully connected neural network, every input is treated as if it were structurally independent from the others. In contrast, a graph neural network respects the actual molecular topology, so nearby atoms communicate through the graph structure. This makes GNNs a natural choice when chemical structure itself is central to the prediction task.

A novel encoding system for molecules is a graph neural network, where molecular structure defines network topology

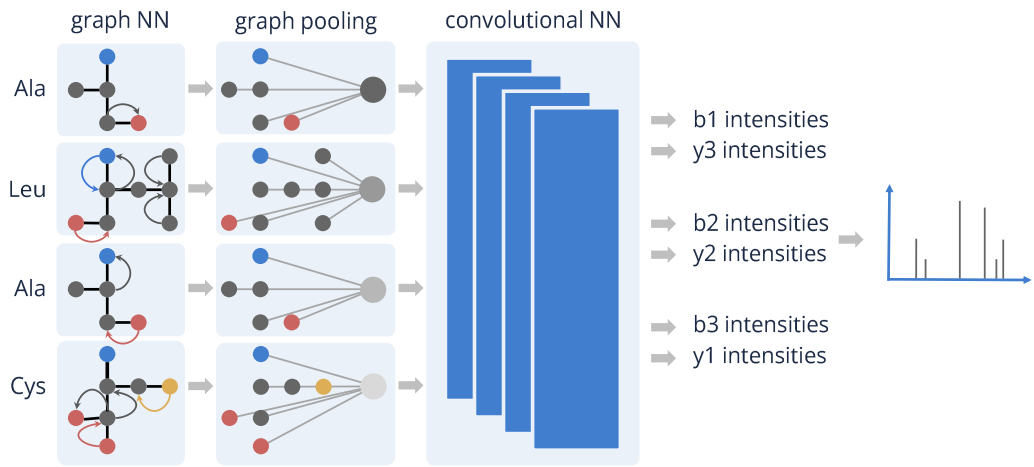




Graph neural network

S Kearnes et al. J Comput Aided Mol Des. (2016) doi:10.1007/s10822-016-9938-8; A Kensert et al. ArXiv (2023) doi:10.48550/arXiv.2208.09944

The graph neural network can be coupled to a convolutional net for peptide spectrum prediction



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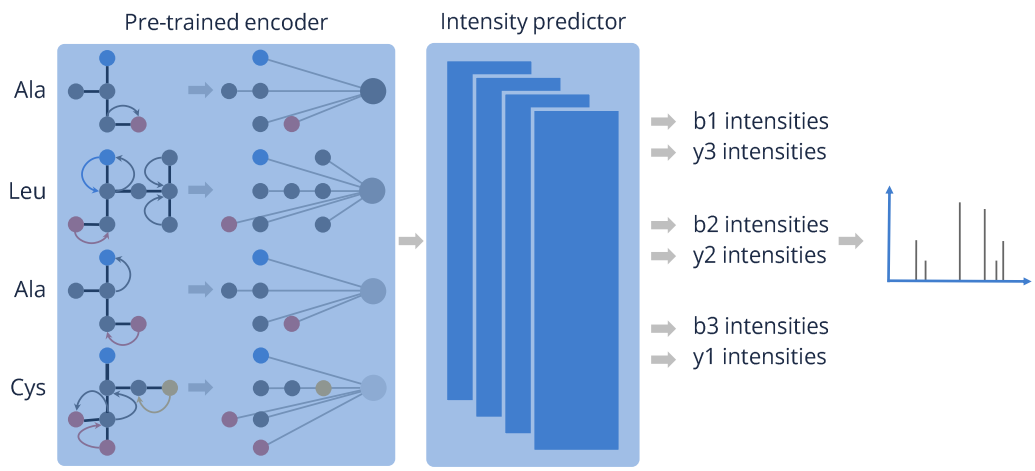
Different neural-network modules can be combined so that each handles the part of the problem it is best suited for. Here, the graph neural network encodes structural information about the molecular building blocks (amino acids), while the convolutional component handles the sequential aspect of the peptide and helps to model the downstream spectrum prediction. This reflects a broader principle in deep learning: complex tasks often benefit from architectures that combine multiple specialized components.

The graph neural network can be pretrained as a foundation model



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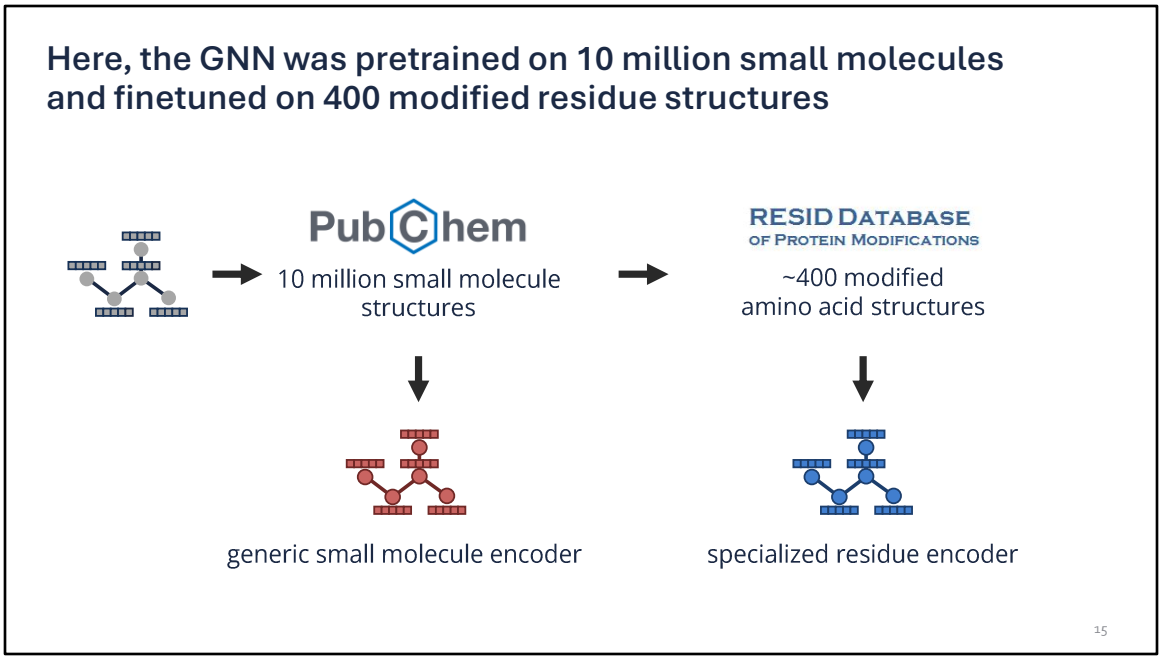
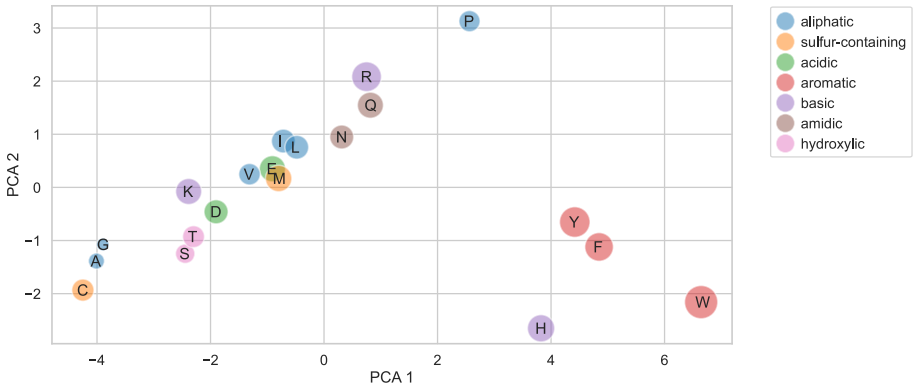
The encoder part of the graph neural network can first be pretrained on a large general molecular dataset before being reused in a more specific downstream model. In that setting, the encoder functions as a **foundation model** for molecular structure. The downstream predictor then benefits from the fact that the encoder already learned broadly useful chemical representations during pretraining.



This example clearly illustrates why pretraining can be so powerful. A model that first sees a very large and diverse molecular dataset can later be adapted to a much smaller specialized dataset that would likely be insufficient for training a strong model from scratch. In other words, large-scale pretraining can compensate, at least partly, for the scarcity of labeled data in a niche biomedical task.

The molecular embedding successfully groups similar amino acids together



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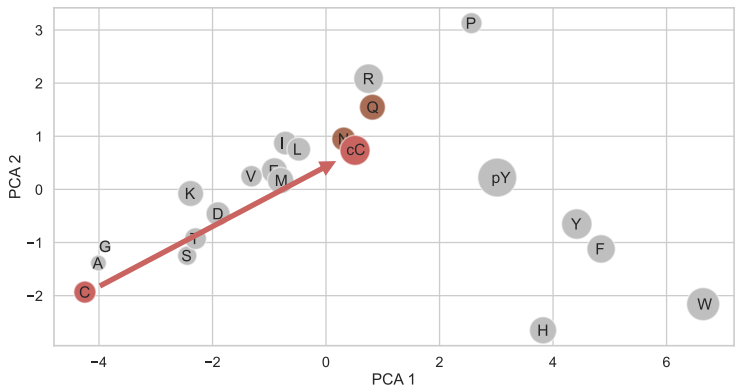
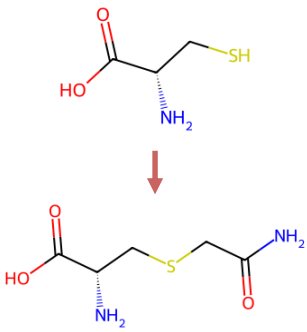
The clustering of similar amino acids suggests that the pretrained model captures meaningful molecular properties. This is one of the central promises of foundation models: they learn **general-purpose representations** that can be reused across tasks.

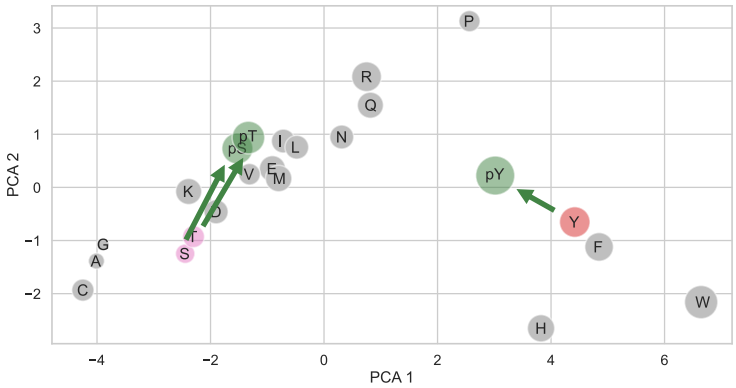
The molecular embedding successfully groups similar amino acids together

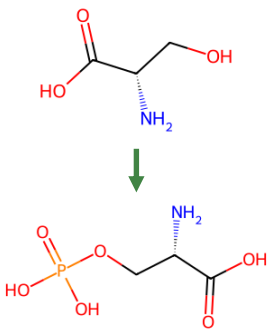




When embeddings organize chemically related molecules together, they provide evidence that the model has learned more than a superficial input-output mapping. Instead, it has formed an internal representation that reflects aspects of the underlying chemistry.

The molecular embedding successfully groups similar amino acids together





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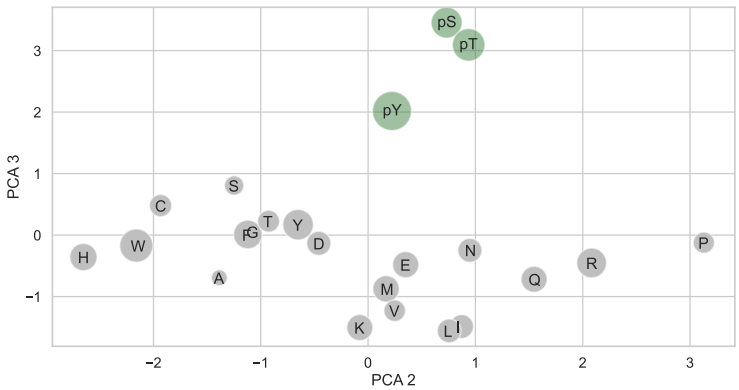

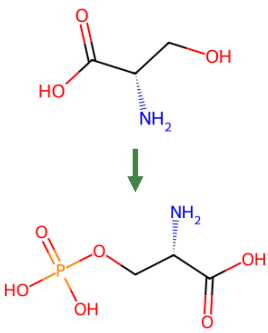

In summary, pretraining and finetuning serve two different goals

	Direct training	Pretraining	Finetuning
Objective	*Learn the 'language' & the specific task at once*	Learn the general 'language' of the training task	Learn the specific training task
Method	*Supervised*	Supervised (with labels) or self-supervised (masking)	Supervised (with labels)
Data requirement	*Large corpus of examples*	Very large corpus of examples	Much smaller data set*
Downstream applicability	*Very specific*	Very diverse	Very specific

* depending on how different the downstream task is from the foundation model.

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Pretraining and fine-tuning are related but distinct stages. **Pretraining** aims to learn the general structure or "language" of the data, often from a large and only weakly labeled or unlabeled dataset. **Fine-tuning** then uses that learned representation to solve a specific downstream task with a smaller, more targeted labeled dataset. This distinction is central to understanding what makes foundation models different from task-specific models trained from scratch.

Recap

- Foundation models first learn the general structure of a data domain from large datasets.
- This pretraining step is often self-supervised, so exact labels are not always required.
- The pretrained model can then be fine-tuned for specific downstream tasks.
- This is very useful in biomedicine, where labeled data are often scarce but unlabeled data can be abundant.